

BACKTESTING EXPECTED SHORTFALL:
A Battery of Eleven Statistics
Across Five Equity Indices and Five Crisis Windows

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Abstract

This paper implements eleven expected-shortfall (ES) and value-at-risk (VaR) backtests on a common out-of-sample slice and reports their joint behaviour on real and simulated data. The battery covers the basic and robust statistics of [Du and Escanciano \(2017\)](#), the parametric Monte-Carlo Z_1 and Z_2 checks of [Acerbi and Szekely \(2014\)](#), the multinomial VaR test of [Kratz et al. \(2018\)](#), the conditional-coverage decomposition of [Christoffersen \(1998\)](#), the proportion-of-failures test of [Kupiec \(1995\)](#), and the standardised-exceedance bootstrap of [McNeil and Frey \(2000\)](#). On the S&P 500 the eleven tests give inconsistent verdicts: nine reject on the Global Financial Crisis window, ten on the COVID-19 window, and the three tests that fail to reject (Christoffersen independence, Acerbi–Szekely Z_1 , McNeil–Frey) do so on both crises. The same pattern holds across a grid of five equity indices and five crisis windows: the unconditional block (U_{ES} , U_{VaR} , Kupiec, multinomial, Z_2) rejects in 19–20 of 20 non-SVB cells, while the day-to-day independence component rejects only 2 of 20. A Monte Carlo size-and-power study on 16,000 simulated paths shows that the marginal-coverage tests over-reject at $n = 1000$ (empirical size up to 16%) and that, under strong asymmetric leverage, Z_2 rejects almost twice as often as Z_1 (0.23 versus 0.12). An LSTM-Quantile baseline trained on 2010–2019 fails to scale during the March 2020 spike, under-calling VaR by a factor of 2.6 relative to AR(1)-GARCH(1,1)- t and rejecting the Kupiec test with a statistic almost twenty times the GARCH value; a mixture-density extension of the same LSTM backbone supplies the analytic CDF the LSTM-Quantile lacks and unlocks the full battery on the neural baseline, reducing the under-call factor from 2.6 to 1.3. A separate four-variant comparison shows that switching the GARCH(1,1) leverage term flips the verdict on five of the eleven tests on the same OOS data. The paper concludes that ES should not be backtested by a single statistic, and that the choice of forecasting model is part of the backtest, not a separable input.

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1 Introduction

The Fundamental Review of the Trading Book (Basel Committee on Banking Supervision, 2019) replaced value-at-risk with expected shortfall as the primary market-risk measure for regulatory capital. VaR at level α is the $(1 - \alpha)$ -quantile of the loss distribution; ES at the same level is the average loss conditional on being beyond the VaR. ES corrects the two well-known shortcomings of VaR: it is sub-additive, hence a coherent risk measure in the sense of Artzner et al. (1999), and it accounts for the magnitude of the losses in the tail rather than their mere frequency.

The transition exposes a tension. VaR admits a simple backtest based on counting violation days, codified in Kupiec (1995) and refined into a conditional-coverage decomposition by Christoffersen (1998). ES does not: Gneiting (2011) shows that ES is not *elicitable* as a single functional, which rules out a strictly proper scoring rule of the kind that exists for VaR (the pinball loss). The literature has responded with a dozen alternative test statistics, each making different assumptions on the data-generating process, on parameter-estimation uncertainty, and on the joint elicitation of (VaR, ES) (Fissler and Ziegel, 2016; Nolde and Ziegel, 2017). Practitioners typically pick one and ship it. The question this paper takes seriously is whether that choice is benign.

Contributions. The paper makes five contributions. First, it implements eleven of the leading ES and VaR backtests in a single unified battery and reports them on the same out-of-sample slice. The battery covers the four basic and the two robust statistics of Du and Escanciano (2017); the parametric Monte-Carlo Z_1 and Z_2 of Acerbi and Szekely (2014); the multinomial test of Kratz et al. (2018); the proportion-of-failures test of Kupiec (1995) and the conditional-coverage decomposition of Christoffersen (1998); and the standardised-exceedance bootstrap of McNeil and Frey (2000). Second, it runs the battery on a grid of five equity indices (S&P 500, Nasdaq 100, FTSE 100, DAX, Nikkei 225) and five crisis windows (Dotcom, the Global Financial Crisis, COVID-19, the 2022 inflation shock, and the March 2023 banking shock) under a common AR(1)-GARCH(1,1)- t forecasting model, producing 275 p-values. Third, it quantifies the finite-sample size and power of the battery on simulated paths drawn from a correctly-specified DGP and from three alternative DGPs with asymmetric leverage and Gaussian tails. Fourth, it isolates the forecasting-model contribution by running the eleven base tests on the four $(\text{vol}, \text{dist}) \in \{\text{GARCH}, \text{GJR}\} \times \{t, \mathcal{N}\}$ combinations on the same OOS window, showing that the GARCH-variant choice flips five of the eleven verdicts. Fifth, it extends the LSTM-Quantile baseline into a mixture-density network whose parametric output unlocks the full battery on the neural model.

Roadmap. Section 2 fixes notation and presents the eleven backtests with their asymptotic null distributions. Section 3 describes the data and the crisis windows. Section 4 reports the empirical results. Section 5 discusses what the battery says about the choice of a single test in operational use. Section 6 concludes. Two appendices document the software architecture and the exact recipe to reproduce every number in the paper.

2 Methodology

Let Y_t denote the percentage log-return at date t , with the sign convention that a loss is $-Y_t > 0$. Let $\mathcal{F}_{t-1}$ denote the information set available through $t - 1$, and let F_t be the one-step-ahead conditional distribution of Y_t given $\mathcal{F}_{t-1}$. Value-at-risk and expected shortfall at level $\alpha \in (0, 1)$ are defined by

$$\text{VaR}_t(\alpha) = -F_t^{-1}(\alpha), \quad \text{ES}_t(\alpha) = \frac{1}{\alpha} \int_0^\alpha \text{VaR}_t(u) du. \quad (1)$$

Under correct conditional specification $\mathbb{P}(-Y_t > \text{VaR}_t(\alpha) | \mathcal{F}_{t-1}) = \alpha$, and ES is the conditional mean of $-Y_t$ given $-Y_t > \text{VaR}_t(\alpha)$.

2.1 PIT residuals and violation indicators

The *probability integral transform* of Y_t under F_t is

$$u_t = F_t(Y_t). \quad (2)$$

Under correct conditional specification $\{u_t\}$ is i.i.d. uniform on $[0, 1]$, which is the substrate the [Du and Escanciano \(2017\)](#) battery exploits. Two derived series carry the test signal:

$$h_t(\alpha) = \mathbf{1}\{u_t \leq \alpha\}, \quad (3)$$

$$H_t(\alpha) = \frac{1}{\alpha} (\alpha - u_t) \mathbf{1}\{u_t \leq \alpha\}. \quad (4)$$

h_t is the binary VaR violation indicator; H_t is its continuous counterpart, equal to zero on a non-violation day and growing linearly with how far in the tail the loss lies on a violation day. Under correct specification,

$$\mathbb{E}[h_t] = \alpha, \quad \text{Var}(h_t) = \alpha(1 - \alpha), \quad \mathbb{E}[H_t] = \frac{\alpha}{2}, \quad \text{Var}(H_t) = \alpha\left(\frac{1}{3} - \frac{\alpha}{4}\right). \quad (5)$$

Two conventions. [Du and Escanciano \(2017\)](#) report ES at $\alpha_{ES} = 2\alpha_{VaR}$ (e.g. 0.10 and 0.05); the Basel III standard ([Basel Committee on Banking Supervision, 2019](#)) sets $\alpha_{VaR} = 0.025$ and keeps the same level for ES. The [Acerbi and Szekely \(2014\)](#) and [McNeil and Frey \(2000\)](#) statistics assume $\alpha_{ES} = \alpha_{VaR}$ because both reference the same set of VaR exceptions. The implementation keeps both conventions in scope: the Du–Escanciano block runs at the paper’s $\alpha_{ES} = 2\alpha_{VaR}$, and the Acerbi–Szekely / McNeil–Frey block re-evaluates the VaR and ES paths at α_{VaR} before running the test, so the legend of every result table reports the exact level used.

2.2 Eleven backtests

The eleven backtests split into three groups. Group A (Du–Escanciano) operates on u_t , h_t and H_t . Group B (VaR baselines) operates on h_t only. Group C (ES baselines) operates on the realised losses, the VaR path, the ES path, and the conditional volatility. Table 1 summarises the asymptotic null distributions. The two robust Du–Escanciano variants (MU_{ES} and MC_{ES}) require the fitted parameter vector, the in-sample window, and the realised series, so they

augment the four basic statistics when those inputs are available.

Group A: Du–Escanciano (2017)

The unconditional ES statistic is

$$U_{ES} = \frac{\sqrt{n} (\bar{H} - \alpha/2)}{\sqrt{\alpha (1/3 - \alpha/4)}}, \quad \bar{H} = \frac{1}{n} \sum_{t=1}^n H_t(\alpha), \quad (6)$$

asymptotically $\mathcal{N}(0, 1)$ under correct specification. The conditional ES Box–Pierce statistic stacks the first m sample autocorrelations $\hat{\rho}_j$ of $H_t - \alpha/2$:

$$C_{ES}(m) = n \sum_{j=1}^m \hat{\rho}_j^2, \quad (7)$$

asymptotically $\chi^2(m)$. U_{VaR} and $C_{VaR}(m)$ are defined identically with $h_t - \alpha$ in place of $H_t - \alpha/2$.

The robust variants correct the variance of the test statistic for parameter-estimation uncertainty. Let $\hat{\theta}$ denote the maximum-likelihood estimator on an in-sample window of length T and let $\hat{W} = T \cdot \widehat{\text{Cov}}(\hat{\theta})$. With

$$\mathbf{R}_{ES} = \left. \frac{\partial \bar{H}}{\partial \theta} \right|_{\hat{\theta}} \quad (8)$$

obtained by central differences (step $\max(|\hat{\theta}_k| \varepsilon_{\text{rel}}, \varepsilon_{\text{min}})$ with $\varepsilon_{\text{rel}} = 10^{-5}$ and $\varepsilon_{\text{min}} = 10^{-7}$), the robust unconditional ES statistic is

$$MU_{ES} = \frac{\sqrt{n} (\bar{H} - \alpha/2)}{\sqrt{\alpha (1/3 - \alpha/4) + (n/T) \mathbf{R}_{ES}^T \hat{W} \mathbf{R}_{ES}}}. \quad (9)$$

$MC_{ES}(m)$ is the analogous correction of $C_{ES}(m)$.

Group B: VaR baselines

Let $N = \sum_t h_t(\alpha)$ be the realised number of violations. The Kupiec proportion-of-failures statistic is

$$\text{LR}_{POF} = -2 \log \left[\frac{\alpha^N (1 - \alpha)^{n-N}}{\hat{\pi}^N (1 - \hat{\pi})^{n-N}} \right], \quad \hat{\pi} = N/n, \quad (10)$$

asymptotically $\chi^2(1)$. The [Christoffersen \(1998\)](#) decomposition adds an independence component LR_{IND} from a two-state Markov chain on $\{h_t\}$ (also $\chi^2(1)$) and a joint conditional-coverage component

$$\text{LR}_{CC} = \text{LR}_{POF} + \text{LR}_{IND} \sim \chi^2(2). \quad (11)$$

The decomposition is exact, so a rejection of LR_{CC} combined with a non-rejection of LR_{IND} identifies the coverage component as the source.

The [Kratz et al. \(2018\)](#) multinomial test takes a vector of K VaR levels ($\alpha_1 < \dots < \alpha_K$) and tests jointly whether the observed counts in the $K + 1$ bins of the empirical distribution match the multinomial model implied by the conditional distribution. The implementation uses $K = 4$ at $\alpha \in \{0.025, 0.05, 0.075, 0.10\}$; the asymptotic null is $\chi^2(K)$.

Group C: ES baselines

The [Acerbi and Szekely \(2014\)](#) Z_1 and Z_2 statistics compare realised losses to the conditional ES:

$$Z_1 = \frac{1}{N} \sum_{t: -Y_t > \text{VaR}_t} \frac{-Y_t}{\text{ES}_t} - 1, \quad (12)$$

$$Z_2 = \sum_{t=1}^n \frac{(-Y_t) \mathbf{1}\{-Y_t > \text{VaR}_t\}}{n \alpha \text{ES}_t} - 1. \quad (13)$$

Z_1 averages the loss-to-ES ratio over the empirical violation set; Z_2 aggregates over the full window. Both have $\mathbb{E}[Z] = 0$ under correct ES specification. The implementation draws n_{sim} paths from the fitted AR(1)-GARCH(1,1)- t at the same conditional moments, replaces the realised innovations with fresh i.i.d. standardised- t draws, and reads the upper-tail Monte-Carlo p-value. $n_{\text{sim}} = 10,000$ in the CLI runs and 2,000 in the Monte-Carlo size-and-power study.

The [McNeil and Frey \(2000\)](#) statistic forms the standardised exceedances

$$r_t = \frac{-Y_t - \text{ES}_t}{\sigma_t} \quad \text{for } t \text{ with a VaR violation}, \quad (14)$$

and tests $\mathbb{E}[r_t \mid -Y_t > \text{VaR}_t] = 0$ against a stationary block bootstrap distribution. Under correct ES specification at the same level as the VaR, the residual of the realised loss against the conditional ES averages to zero on the violation set, since ES is precisely the conditional mean of the loss given a violation. The CLI default is $n_{\text{boot}} = 10,000$.

Table 1: The eleven-test battery. The robust statistics MU_{ES} and MC_{ES} require the fitted parameter vector and the in-sample window; the basic battery runs without them.

Test	Targets	Asymptotic null	Reference
U_{ES}	ES unconditional	$\mathcal{N}(0, 1)$	Du and Escanciano 2017
$C_{ES}(m)$	ES conditional (Box–Pierce)	$\chi^2(m)$	Du and Escanciano 2017
MU_{ES}	ES unconditional, robust	$\mathcal{N}(0, 1)$	Du and Escanciano 2017
$MC_{ES}(m)$	ES conditional, robust	$\chi^2(m)$	Du and Escanciano 2017
U_{VaR}	VaR unconditional	$\mathcal{N}(0, 1)$	Du and Escanciano 2017
$C_{\text{VaR}}(m)$	VaR conditional	$\chi^2(m)$	Du and Escanciano 2017
Kupiec_{POF}	VaR proportion-of-failures	$\chi^2(1)$	Kupiec 1995
$\text{Christoffersen}_{CCI}$	VaR independence	$\chi^2(1)$	Christoffersen 1998
$\text{Christoffersen}_{CC}$	VaR coverage + independence	$\chi^2(2)$	Christoffersen 1998
$\text{KLM}_{\text{mult}}(K)$	ES via multi-level VaR	$\chi^2(K)$	Kratz et al. 2018
Z_1, Z_2	ES (parametric Monte Carlo)	simulated	Acerbi and Szekely 2014
McNeil–Frey	ES via standardised exceedances	bootstrap	McNeil and Frey 2000

2.3 Forecasting model

The baseline conditional model is AR(1)-GARCH(1,1) with standardised Student- t innovations:

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad (15)$$

$$\varepsilon_t = \sigma_t z_t, \quad (16)$$

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad (17)$$

$$z_t \stackrel{\text{i.i.d.}}{\sim} t_\nu \text{ standardised to unit variance.} \quad (18)$$

The parameters $(c, \phi, \omega, \alpha_1, \beta_1, \nu)$ are estimated by conditional maximum likelihood using the arch package (Engle, 1982; Bollerslev, 1986). The recursion is run on the full history so the out-of-sample residuals are conditioned on the in-sample-fitted dynamics rolling forward without re-estimation; the recursion clips σ_t^2 to be non-negative before the square root to absorb numerical noise from ill-conditioned MLE fits on very short windows.

Two variants extend the baseline: the GJR-GARCH(1,1,1) of Glosten et al. (1993),

$$\sigma_t^2 = \omega + (\alpha_1 + \gamma \mathbf{1}\{\varepsilon_{t-1} < 0\}) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad (19)$$

which adds the asymmetric leverage term γ , and a Gaussian innovation flavour where $z_t \sim \mathcal{N}(0, 1)$. The four combinations (GARCH, t), (GARCH, $\mathcal{N}$), (GJR, t), (GJR, $\mathcal{N}$) share a single recursion helper with $\gamma = 0$ recovering the symmetric case.

Two neural baselines are included for contrast. The first is the *LSTM-Quantile* head of Hochreiter and Schmidhuber (1997), which maps a length- L window of past returns to two outputs ($\hat{\text{VaR}}_t, \hat{\text{ES}}_t$) through a stacked LSTM whose hidden state feeds two linear heads. The ES head is parameterised as $\hat{\text{ES}}_t = \hat{\text{VaR}}_t + \text{softplus}(\cdot)$, so $\hat{\text{ES}}_t \geq \hat{\text{VaR}}_t$ holds by construction. The training loss is the pinball loss for VaR plus the FZ0 loss of Fissler and Ziegel (2016) for the joint (VaR, ES) pair. The LSTM-Quantile has no closed-form PIT and so receives only the subset of the battery that does not require u_t .

The second is the *mixture-density LSTM* (MDN-LSTM), which shares the same LSTM backbone but replaces the two-output head with three linear heads that parameterise the conditional Student- t on the next return: $Y_{t+1} \sim t_{\nu_t}(\mu_t, \sigma_t)$. The scale σ_t is positivity-constrained via softplus and the degrees of freedom ν_t are constrained above two (so the variance is finite) via softplus($\cdot$) + 2. The training loss is the negative log-likelihood of the predicted distribution, which is a strictly proper scoring rule. Because the MDN supplies a closed-form CDF, every quantity the battery needs follows in closed form: the PIT residual $u_t = \text{cdf}_{t_{\nu_t}}(Y_t \mid \mu_t, \sigma_t)$, the analytic VaR and ES, and a per-day Student- t simulator for the Acerbi-Szekely Monte-Carlo p -values. Eleven of the thirteen battery statistics therefore apply on the MDN-LSTM; the two robust Du-Escanciano variants MU_{ES} and MC_{ES} remain unreachable because they require the parameter covariance of an MLE fit, which the stochastic-gradient-trained network does not expose in closed form.

3 Data

The data are end-of-day adjusted closing prices for five equity indices, downloaded from Yahoo Finance and converted to log-returns in percent. Table 2 lists the tickers and the in-sample and out-of-sample windows. The full S&P 500 sample covers 7,044 trading days from January 1997 to December 2024 and is pinned to a frozen parquet file under `tests/data/` so back-fills by the upstream vendor cannot alter the reference numbers in this paper.

Table 2: Equity indices and crisis windows. Each (asset, crisis) cell uses a five-year in-sample window ending the day before the crisis out-of-sample slice.

Crisis	Out-of-sample window	Trading days	In-sample window
Dotcom	2000-04-01 – 2002-12-31	689	1995-04-01 – 2000-03-31
Global Financial Crisis	2007-07-01 – 2009-06-30	504	2002-07-01 – 2007-06-30
COVID-19	2020-01-01 – 2021-12-31	505	2015-01-01 – 2019-12-31
Inflation 2022	2022-01-01 – 2022-12-31	251	2017-01-01 – 2021-12-31
SVB / regional banks	2023-03-01 – 2023-09-30	148	2018-03-01 – 2023-02-28

Asset	Ticker	Market
S&P 500	^GSPC	US large cap
Nasdaq 100	^NDX	US tech
FTSE 100	^FTSE	UK large cap
DAX	^GDAXI	German large cap
Nikkei 225	^N225	Japan large cap

The single-asset replication in Section 4.1 uses the S&P 500 over two of these crises (GFC and COVID-19), with the in-sample windows from [Du and Escanciano \(2017\)](#) extended by one decade. The multi-asset grid in Section 4.2 uses all five (asset, crisis) combinations on the same five-year-in-sample template.

4 Empirical results

4.1 Single-asset replication: S&P 500, GFC and COVID-19

Setup. Risk levels follow the [Du and Escanciano \(2017\)](#) convention: $\alpha_{VaR} = 0.05$, $\alpha_{ES} = 0.10$, Box–Pierce lag $m = 5$. Two periods are reported: the Global Financial Crisis with an in-sample window covering 1997-01-01 to 2007-06-30 ($T = 2,640$ days) and an OOS slice of $n = 504$ days from 2007-07-01 to 2009-06-30; and the COVID-19 crash with in-sample 2010-01-01 to 2019-12-31 ($T \approx 2,516$ days) and OOS $n = 505$ days from 2020-01-01 to 2021-12-31. The AR(1)-GARCH(1,1)- t fit gives $\hat{v} = 8.72$ on the GFC in-sample window and $\hat{v} = 4.83$ on the COVID in-sample window. The 2010–2019 in-sample tail is empirically fatter than the 1997–2007 in-sample tail; neither training window contains the March 2020 shock, which lies entirely in the COVID OOS slice. Realised VaR violations are 44 out of 504 on the GFC OOS slice and 40 out of 505 on the COVID OOS slice; both exceed the 5% nominal rate of 25 violations.

Battery results. Table 3 reports the full battery on both windows: the eleven base tests plus the two robust Du–Escanciano variants MU_{ES} and MC_{ES} , for a total of thirteen statistics per

period. Nine tests reject at the 5% level on the GFC window; ten reject on COVID-19. The four tests that fail to reject on at least one of the two crises are $C_{VaR}(5)$ on GFC ($p = 0.0894$), Christoffersen_{CCI} on both ($p = 0.0645$ on GFC, 0.2979 on COVID), Z_1 on both ($p = 0.2168$ on GFC, 0.1021 on COVID), and McNeil–Frey on both ($p = 0.2759$ on GFC, 0.1721 on COVID).

Table 3: Single-asset battery: thirteen statistics on the S&P 500 over the Global Financial Crisis (in-sample 1997–2007, OOS 2007-07 to 2009-06) and COVID-19 (in-sample 2010–2019, OOS 2020-01 to 2021-12). Rejection at 5% in bold.

Test	GFC 2008		COVID-19		Decision	
	Statistic	p-value	Statistic	p-value	GFC	COVID
U_{ES}	4.1968	0.0000	2.3129	0.0207	R	R
$C_{ES}(5)$	14.9532	0.0106	13.8442	0.0166	R	R
U_{VaR}	3.8423	0.0001	3.0116	0.0026	R	R
$C_{VaR}(5)$	9.5390	0.0894	17.0723	0.0044	F	R
MU_{ES}	3.7395	0.0002	2.2049	0.0275	R	R
$MC_{ES}(5)$	14.8936	0.0108	12.6821	0.0265	R	R
Kupiec _{POF}	12.1945	0.0005	7.7625	0.0053	R	R
Christoffersen _{CCI}	3.4167	0.0645	1.0836	0.2979	F	F
Christoffersen _{CC}	15.6112	0.0004	8.8461	0.0120	R	R
KLM _{mult} (4)	23.3104	0.0001	13.5039	0.0091	R	R
Z_1	0.0380	0.2168	0.0991	0.1021	F	F
Z_2	0.8124	0.0002	0.7412	0.0004	R	R
McNeil–Frey	1.0796	0.2759	1.3609	0.1721	F	F
Rejections at 5%					9/13	10/13

The marginal-coverage block (U_{ES} , U_{VaR} , Kupiec_{POF}, KLM_{mult}(4)) and the conditional-coverage block ($C_{ES}(5)$, MU_{ES} , $MC_{ES}(5)$, plus $C_{VaR}(5)$ on COVID) fire on both crises. The day-to-day independence test Christoffersen_{CCI} and the empirical-violation-set statistics Z_1 and McNeil–Frey do not. The asymmetry is operational: the model over-violates on aggregate but does not concentrate the excess violations on consecutive days. The robust Du–Escanciano variants MU_{ES} and $MC_{ES}(5)$ land within 0.5 standard errors of their basic counterparts on both windows, indicating that the parameter-estimation correction is small at the present $n/T \approx 0.2$ ratio.

Figure 1 shows the realised returns against the negative VaR and ES envelopes on the GFC OOS slice; Figure 2 shows the resulting h_t and H_t series. Figure 3 reports the PIT residual diagnostic on the COVID window: the Kolmogorov–Smirnov test against Uniform(0, 1) returns $p = 0.0929$, so the marginal distribution of u_t on COVID is consistent with correct specification at 5% even though the joint hypothesis is rejected by ten of the thirteen tests in Table 3. This is precisely the kind of contrast the battery is designed to expose: marginal-quantile fit can look acceptable while the conditional dynamics fail at the tail.

4.2 Multi-asset, multi-crisis grid

The grid evaluates the basic eleven-test battery on each of the 25 (asset, crisis) cells. The robust variants MU_{ES} and MC_{ES} are absent from the grid because the multi-asset harness does not propagate the fitted parameter vector and the realised in-sample series required for the central-difference Jacobian. Every cell estimates without numerical issues; the grid totals

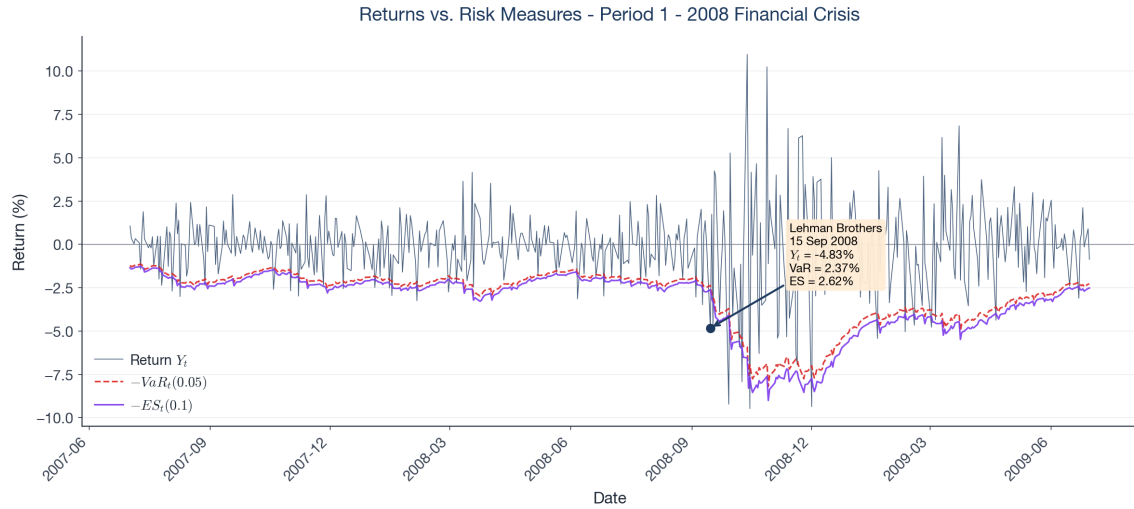


Figure 1: S&P 500 daily returns and one-step-ahead negative VaR and ES envelopes over the Global Financial Crisis out-of-sample window ($\alpha_{VaR} = 0.05$, $\alpha_{ES} = 0.10$). The annotation identifies the 15 September 2008 Lehman Brothers collapse, where the realised return (-4.83%) exceeded both the VaR (2.37%) and the ES (2.62%) envelopes.

$5 \times 5 \times 11 = 275$ p-values.

Table 4 reports the rejection rate of each test across the five assets, period by period. The rejection rate is the share of assets (out of five) that reject the null at the 5% level on the given crisis window. The patterns observed on the S&P 500 in Section 4.1 hold across the four pre-2023 windows: the marginal-coverage block reaches near-unanimous rejection, Christoffersen_{CCI} stays at or below 20%, and Z_2 dominates Z_1 by a wide margin.

Table 4: Rejection rate per (test, crisis) cell. Each entry is the share of assets (out of five) that reject the test at the 5% level on the given crisis window. The right-most column totals the four pre-2023 crises (20 cells per test) for compactness.

Test	COVID	Dotcom	GFC	Inflation 2022	SVB	Pre-SVB rej./20
U_{ES}	1.00	1.00	1.00	1.00	0.00	20
$C_{ES}(5)$	0.40	0.40	0.60	0.40	0.00	9
U_{VaR}	0.80	1.00	1.00	1.00	0.00	19
$C_{VaR}(5)$	0.20	0.40	0.40	0.40	0.00	7
Z_1	0.40	0.00	0.60	0.00	0.00	5
Z_2	1.00	0.80	1.00	1.00	0.20	19
McNeil–Frey	0.40	0.20	0.60	0.20	0.20	7
Kupiec _{POF}	0.80	1.00	1.00	1.00	0.00	19
Christoffersen _{CCI}	0.00	0.00	0.20	0.20	0.00	2
Christoffersen _{CC}	0.60	1.00	1.00	1.00	0.00	18
KLM _{mult} (4)	1.00	1.00	1.00	1.00	0.00	20

Two patterns stand out. First, the SVB column is quiet: of the 55 test–asset cells on the SVB window, only 2 reject (one Z_2 , one McNeil–Frey). The 2023 banking shock saw sharp moves in regional bank equities but limited index-level volatility, so the AR(1)-GARCH(1,1)- t scales its envelope correctly. This is a useful sanity check: a window labelled “crisis” does not have to coincide with ES mis-specification on diversified indices.

Second, the gap between Christoffersen_{CCI} (2/20 rejections outside SVB) and Christoffersen_{CC}

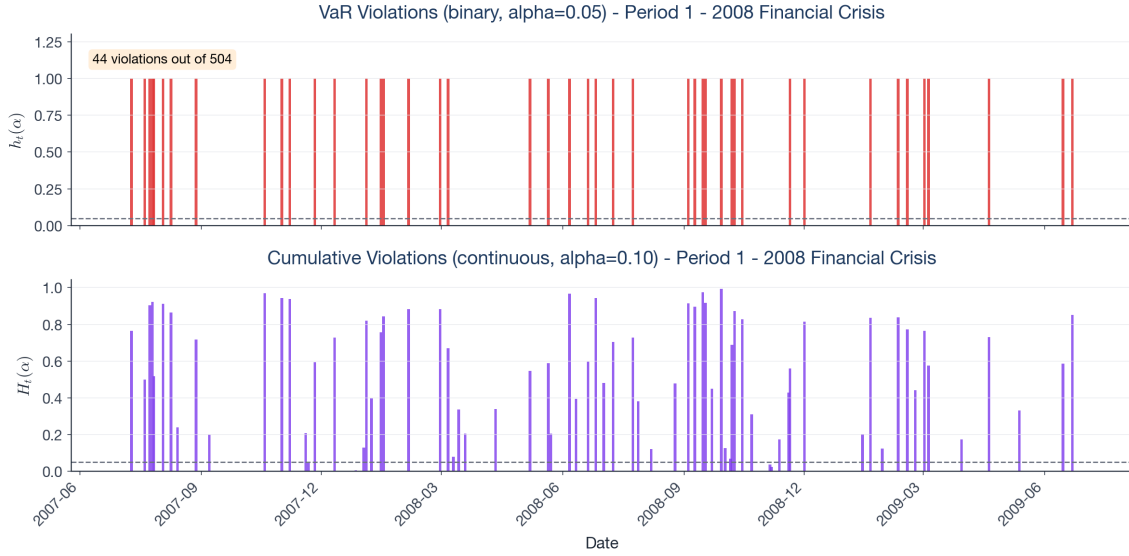


Figure 2: Violation indicators over the GFC OOS window. Top: binary VaR violation indicator $h_t = \mathbf{1}\{u_t \leq \alpha_{VaR}\}$, with the horizontal line at the nominal rate $\alpha_{VaR} = 0.05$. Bottom: cumulative ES severity $H_t = \alpha_{ES}^{-1}(\alpha_{ES} - u_t) \mathbf{1}\{u_t \leq \alpha_{ES}\}$, with the horizontal line at the expectation $\alpha_{ES}/2$ under correct specification.

(18/20) is carried almost entirely by LR_{POF} , which on its own rejects 19/20 cells. The decomposition $LR_{CC} = LR_{POF} + LR_{IND}$ is exact, and the empirical evidence assigns the rejection signal to the marginal-coverage component. The day-to-day independence component contributes a single extra rejection on the GFC and on the inflation window. Pooling the two components into LR_{CC} thus inflates the apparent power of the joint test, but the source of the rejection is the violation count, not the clustering pattern.

Figure 4 renders the full 11×25 p-value grid; Figure 5 aggregates over the 25 cells to give a per-test summary.

4.3 Monte Carlo size-and-power study

The Monte Carlo study evaluates the finite-sample behaviour of each test under four scenarios:

- $H_0^{(t)}$: AR(1)-GARCH(1,1)- t with $\nu = 8.0$ (correctly specified).
- $H_1^{(\gamma=0.15)}$: GJR-GARCH(1,1,1)- t with $\gamma = 0.15$ (moderate asymmetric leverage).
- $H_1^{(\gamma=0.25)}$: GJR-GARCH(1,1,1)- t with $\gamma = 0.25$ (strong asymmetric leverage).
- $H_1^{(N)}$: AR(1)-GARCH(1,1)- $\mathcal{N}$ (Gaussian innovations, same volatility dynamics).

For each scenario and each sample size $n \in \{250, 500, 1,000, 2,500\}$ the study draws 1,000 replications, fits an AR(1)-GARCH(1,1)- t on the first 60% of each path, and runs the basic eleven-test battery on the remaining 40%. The Z_1 and Z_2 p-values are parametric Monte-Carlo readings from a StandardizedTSimulator with 2,000 draws per replication; the McNeil–Frey bootstrap uses 200 block-bootstrap samples per replication in the Monte-Carlo study (vs. 10,000 in the CLI runs) to keep the grid affordable. The full size-and-power log contains 176,000 rows; the snapshot at $n = 1,000$ is reported in Table 5.

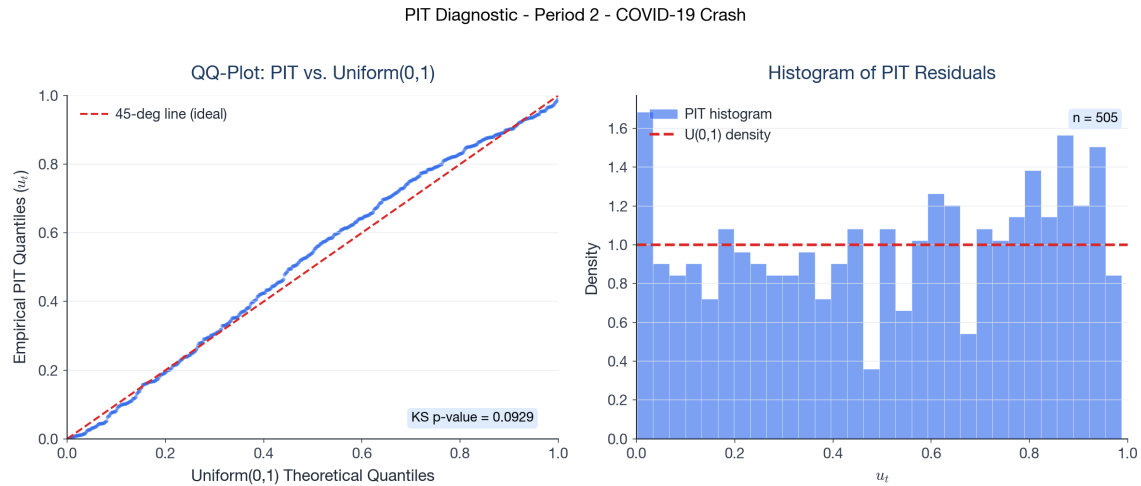


Figure 3: PIT residual diagnostic for the COVID-19 OOS window. Left: QQ-plot of the sorted u_t against uniform quantiles, with the 45-degree reference line. Right: histogram of u_t against the uniform density. The Kolmogorov–Smirnov test against Uniform(0, 1) returns $p = 0.0929$, so the marginal distribution of u_t is consistent with correct specification even though six of the eleven backtests reject the joint hypothesis at 5%.

Table 5: Empirical rejection rate at $n = 1,000$. Columns: rejection rate under the correctly-specified DGP (H_0) and under the three alternatives. Values in bold sit at or above 25%; values in italics deviate from the nominal 5% by more than 4 pp under H_0 (size distortion).

Test	H_0 : GARCH- t	H_1 : GJR $\gamma=.15$	H_1 : GJR $\gamma=.25$	H_1 : Normal
U_{ES}	0.16	0.17	0.31	0.17
$C_{ES}(5)$	0.08	0.16	0.30	0.09
U_{VaR}	0.14	0.13	0.28	0.15
$C_{VaR}(5)$	0.08	0.14	0.24	0.07
Z_1	0.12	0.10	0.12	0.07
Z_2	0.14	0.14	0.23	0.12
McNeil–Frey	0.16	0.16	0.14	0.14
Kupiec $_{POF}$	0.13	0.14	0.26	0.15
Christoffersen $_{CCI}$	0.04	0.05	0.10	0.04
Christoffersen $_{CC}$	0.12	0.13	0.26	0.11
KLM $_{mult}(4)$	0.13	0.15	0.25	0.13

Under the correctly-specified DGP the conditional Box–Pierce tests $C_{ES}(5)$ and $C_{VaR}(5)$ (0.08) and the day-to-day independence test Christoffersen $_{CCI}$ (0.04) are within four percentage points of the nominal 5% size. The remaining eight tests over-reject: U_{ES} and McNeil–Frey at 0.16, U_{VaR} and Z_2 at 0.14, Kupiec $_{POF}$ and KLM $_{mult}(4)$ at 0.13, Z_1 and Christoffersen $_{CC}$ at 0.12. The over-rejection narrows but does not vanish at $n = 2,500$ (Figure 6), consistent with the slow convergence of the asymptotic χ^2 approximations for marginal-coverage tests.

Under the moderate leverage alternative ($\gamma = 0.15$) all rejection rates stay below 20%. Under the strong leverage alternative ($\gamma = 0.25$) the conditional Box–Pierce block leads: U_{ES} at 0.31, $C_{ES}(5)$ at 0.30, U_{VaR} at 0.28, Kupiec $_{POF}$ and Christoffersen $_{CC}$ at 0.26, KLM $_{mult}(4)$ at 0.25, $C_{VaR}(5)$ at 0.24, Z_2 at 0.23. The net power (rejection rate at H_1 minus rejection rate at H_0) is +22 pp for $C_{ES}(5)$, +16 pp for $C_{VaR}(5)$, +15 pp for U_{ES} , +14 pp for U_{VaR} and Christoffersen $_{CC}$, +13 pp for Kupiec $_{POF}$, +12 pp for KLM $_{mult}(4)$, +9 pp for Z_2 , +6 pp for Christoffersen $_{CCI}$, 0 pp for Z_1 , and –2 pp for McNeil–Frey. The conditional Box–Pierce statistics on cumulative violations are the

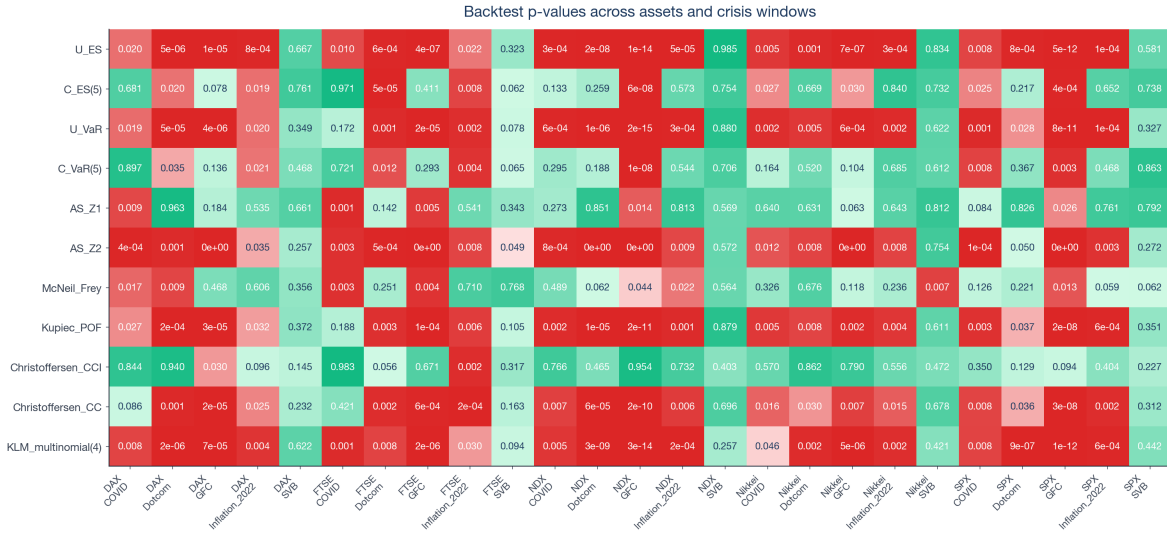


Figure 4: Heatmap of p-values across (test) $\times$ (asset, crisis) cells. Cells with $p < 0.05$ are shaded red, cells failing to reject are shaded green; the intensity tracks distance from the threshold. Eleven tests $\times$ twenty-five cells gives 275 p-values.

most powerful of the eleven tests against leverage misspecification.

The Z_1 versus Z_2 contrast is sharp: at $n = 1,000$, Z_2 has net power +9 pp under strong leverage while Z_1 has 0 pp. The gap reflects that Z_2 aggregates the loss-to-ES ratio over the full OOS window and so picks up an excess number of violations, while Z_1 normalises only over the empirical violation set and remains blind to that channel. Practitioners reaching for “the Acerbi-Szekely test” should be aware that the two variants behave very differently against asymmetric leverage.

Under the Gaussian-innovation alternative every rejection rate stays within 4 pp of its H_0 value, including the three tests specifically designed to track tail mis-specification (Z_1 , Z_2 , McNeil-Frey). At $n = 1,000$ the mismatch between a Gaussian tail and a Student- t tail predicted by the fitted model is not pronounced enough in the PIT residuals to register on any of the u_t -based tests; sharper detection requires a larger n or a heavier-tailed alternative. Power curves over $n \in \{250, 500, 1,000, 2,500\}$ are shown in Figure 7.

4.4 LSTM-Quantile versus AR(1)-GARCH(1,1)- t on COVID-19

The neural baseline compares an LSTM-Quantile head to the AR(1)-GARCH(1,1)- t baseline on the COVID-19 OOS slice. Both models are evaluated at $\alpha = 0.05$. The LSTM is trained on the 2010-2019 S&P 500 in-sample window ($n_{IS} = 2,516$ days) for 30 epochs with sequence length $L = 20$, hidden size 32, batch size 64, Adam at learning rate 10^{-3} , and a joint pinball + FZ0 loss. The training runs on Apple’s MPS accelerator; the final pinball loss is 0.7906 and the final FZ0 loss is 0.8943. Both models emit one-step-ahead ($\text{VaR}_t, \text{ES}_t$) for every OOS day ($n_{OOS} = 505$).

Table 6 reports the mean VaR and ES forecasts over the OOS slice. The LSTM under-calls VaR by a factor of $1.831/0.695 = 2.63$: trained on the calm 2010–2019 decade, it produces a VaR envelope anchored to that regime’s volatility level rather than to the realised regime during the OOS slice. The mean ES values are closer (2.682 vs. 2.839), but the LSTM’s VaR-to-ES spread is much wider than the GARCH’s, an artefact of the softplus parameterisation that decouples the

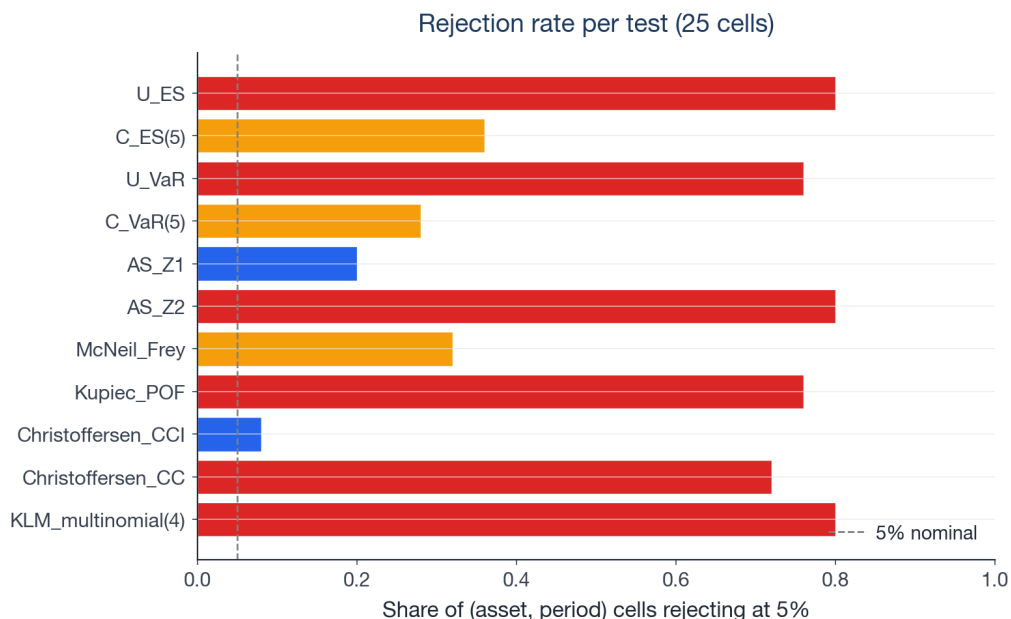


Figure 5: Share of the 25 (asset, crisis) cells in which each test rejects at the 5% level. The vertical line at 5% marks the size a correctly-specified test should achieve under the null.

Table 6: Mean VaR and ES forecasts over the COVID-19 OOS slice. The LSTM trained on 2010–2019 under-calls VaR by a factor of 2.63.

Model	Mean VaR (%)	Mean ES (%)
AR(1)-GARCH(1,1)- t	1.831	2.682
LSTM-Quantile	0.695	2.839

two heads on tail days.

Table 7 reports the subset of the battery that does not require an analytical PIT. Both models reject the Kupiec POF test, but the LSTM’s statistic is 150.636 versus the GARCH’s 7.762, a factor of 19.4. The Christoffersen independence component fails to reject for either model, so the violation rate is elevated on both but the violations are not consecutively clustered in the sense of the two-state Markov chain. The Acerbi-Szekely statistics are reported but their p-values would require a parametric simulator of the LSTM’s forecast distribution, which the softplus-quantile head does not provide; a mixture-density variant would supply the missing PIT and is left as future work.

The visual contrast in Figure 8 makes the point: the GARCH envelope tracks the regime shift in March 2020 by construction, while the LSTM envelope holds approximately constant. The neural baseline does not generalise to a once-in-a-decade tail event when trained on a regime that does not contain one.

4.5 Mixture-density LSTM: closed-form CDF and the full battery

The LSTM-Quantile head produces $(\widehat{\text{VaR}}, \widehat{\text{ES}})$ and nothing else. Six of the thirteen statistics in Table 3 require the PIT residual $u_t = F_t(Y_t)$ or a parametric simulator, which a quantile-only head cannot supply. The mixture-density LSTM (MDN-LSTM) replaces the two-output head with three linear heads on the same LSTM backbone, parameterising the Student- t distribution of

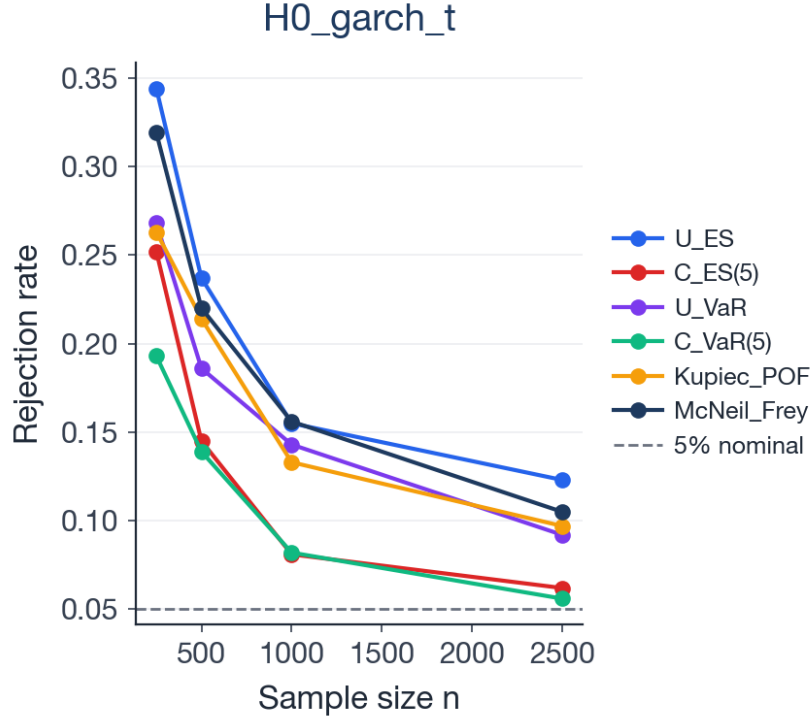


Figure 6: Empirical size of six representative tests under the correctly-specified DGP, as a function of the OOS sample size n . The dashed line marks the nominal 5% level.

Table 7: Battery comparison on the COVID-19 OOS slice. Five tests that do not require an analytical PIT are reported; the Acerbi–Szekely p-values would need a parametric simulator of the LSTM forecast distribution, which is not available.

Test	GARCH- t			LSTM		
	Stat.	p-value	Decision	Stat.	p-value	Decision
Kupiec _{POF}	7.762	0.0053	R	150.636	0.0000	R
Christoffersen _{CCI}	1.084	0.2979	F	0.468	0.4941	F
Christoffersen _{CC}	8.846	0.0120	R	151.103	0.0000	R
Z_1	0.099	N/A	—	−0.348	N/A	—
Z_2	0.741	N/A	—	1.685	N/A	—

the next return $Y_{t+1} \sim t_{v_t}(\mu_t, \sigma_t)$. The training loss is the negative log-likelihood of the predicted distribution. From the trained network, the PIT, the analytic VaR and the analytic ES follow in closed form; a per-day simulator generates fresh Student- t paths for the Acerbi–Szekely Monte-Carlo p-values. The robust Du–Escanciano statistics (MU_{ES} , MC_{ES}) remain unreachable because they require the parameter covariance of an MLE fit, which the stochastic-gradient-trained network does not expose in closed form.

Trained on the same 2010-2019 window as the LSTM-Quantile and GARCH- t baselines, the MDN-LSTM converges to a final NLL of 1.121 and produces an OOS mean VaR of 1.445% over the COVID-19 window. That sits between the LSTM-Quantile’s 0.695% and the GARCH- t ’s 1.831% (Table 8): the MDN scales the envelope substantially better than the LSTM-Quantile but still under-calls relative to GARCH- t . Average conditional parameters on the OOS slice are $\bar{\mu} = 0.11$, $\bar{\sigma} = 0.77$, $\bar{v} = 4.67$ – a fat-tailed regime, consistent with what the GARCH- t fit reports

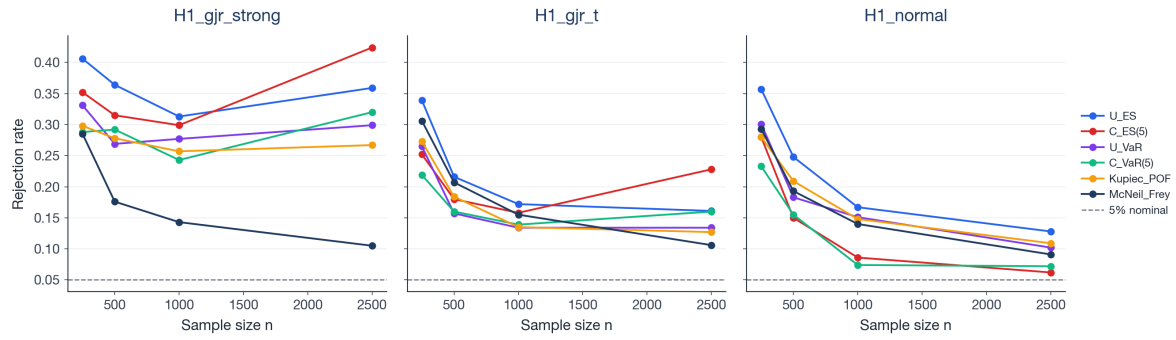


Figure 7: Empirical power curves of six representative tests under the three alternative DGPs (left to right: strong GJR leverage, moderate GJR leverage, Gaussian innovations), as a function of the OOS sample size n . The dashed line marks the nominal 5% level.

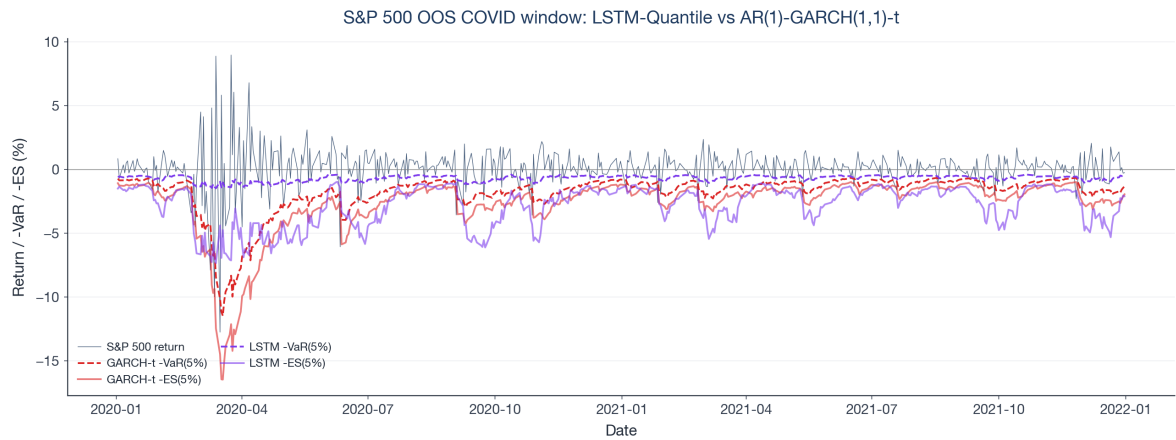


Figure 8: S&P 500 daily returns over the COVID-19 OOS slice (2020–2021) with the negative VaR and ES envelopes of the AR(1)-GARCH(1,1)- t baseline and of the LSTM-Quantile head. The LSTM, trained on the calm 2010–2019 decade, holds its envelope close to that decade’s level and is consistently breached during the March 2020 spike.

on the same window ($\hat{v} = 4.83$).

Table 8: Mean VaR and ES forecasts on the COVID-19 OOS slice, $\alpha = 0.05$. The MDN-LSTM sits between the LSTM-Quantile and the GARCH- t baseline; the LSTM-Quantile under-calls VaR by a factor of 2.63 relative to GARCH- t , the MDN by a factor of 1.27.

Model	Mean VaR (%)	Mean ES (%)
AR(1)-GARCH(1,1)- t	1.831	2.682
LSTM-Quantile	0.695	2.839
MDN-LSTM	1.445	2.136

Table 9 reports the side-by-side battery on the COVID-19 OOS slice. The MDN’s Kupiec POF statistic (21.6) is a factor of 2.8 above the GARCH- t value (7.8). The LSTM-Quantile’s was a factor of 19.4 in Table 7: the parametric output substantially mitigates the under-call, but a calm training window still under-prepares the MDN for a once-in-a-decade tail event. The Acerbi-Szekely Z_1 inverts: GARCH- t fails to reject ($p = 0.102$) where the MDN does ($p = 0.043$). The MDN predicts a smaller ES on the violation days, and the realised severities then exceed it on average – precisely the channel Z_1 is designed to detect.

Table 9: Battery comparison on COVID-19 OOS, $\alpha = 0.05$. The GARCH- t column reports thirteen statistics (including the two robust Du–Escanciano variants); the MDN-LSTM column reports eleven (the robust pair is unavailable). **R**: reject at 5%. **F**: fail to reject.

Test	GARCH- t			MDN-LSTM		
	Stat.	p-value	Dec.	Stat.	p-value	Dec.
U_{ES}	3.302	0.0010	R	5.535	0.0000	R
$C_{ES}(5)$	7.532	0.1840	F	37.767	0.0000	R
MU_{ES}	3.043	0.0023	R	N/A	N/A	—
$MC_{ES}(5)$	5.906	0.3155	F	N/A	N/A	—
U_{VaR}	3.012	0.0026	R	5.258	0.0000	R
$C_{VaR}(5)$	17.072	0.0044	R	24.669	0.0002	R
Z_1	0.099	0.1021	F	0.158	0.0426	R
Z_2	0.741	0.0004	R	1.340	0.0000	R
McNeil–Frey	1.361	0.1721	F	1.502	0.1192	F
Kupiec $_{POF}$	7.762	0.0053	R	21.614	0.0000	R
Christoffersen $_{CCI}$	1.084	0.2979	F	0.006	0.9370	F
Christoffersen $_{CC}$	8.846	0.0120	R	21.620	0.0000	R
KLM $_{mult}(4)$	13.504	0.0091	R	29.583	0.0000	R
Rejections at 5%				8/13	vs.	9/11

4.6 Forecasting-model sensitivity: four GARCH variants

The replication picks AR(1)-GARCH(1,1)- t because that is the forecasting model in [Du and Escanciano \(2017\)](#). The choice is not innocuous: four nearby specifications differ in their backtest verdict on the same COVID-19 OOS window. Table 10 runs the eleven base tests on the four combinations $(\text{vol}, \text{dist}) \in \{\text{GARCH}, \text{GJR}\} \times \{t, N\}$, sharing the AR(1) mean and the GARCH(1,1) recursion and differing only in the leverage term (GJR adds $\gamma \mathbf{1}\{\varepsilon_{t-1} < 0\} \varepsilon_{t-1}^2$) and the innovation distribution.

Table 10: Eleven backtests across four GARCH variants on the S&P 500 COVID-19 OOS slice. Cells in bold are rejections at 5%. The last two columns report the in-sample log-likelihood (2010-2019, $T = 2,516$ days) and the rejection count at 5%. The Acerbi–Szekely cells for the two Normal-innovation variants are empty because the parametric simulator wired into the battery is Student- t -only.

Test	GJR- t	GARCH- t	GJR- N	GARCH- N
U_{ES}	0.0111	0.0199	0.0600	0.0418
$C_{ES}(5)$	0.2366	0.0164	0.4153	0.0385
U_{VaR}	0.0159	0.0025	0.0452	0.0273
$C_{VaR}(5)$	0.9899	0.0045	0.5274	0.0117
Z_1	0.0140	0.1070	—	—
Z_2	0.0000	0.0006	—	—
McNeil–Frey	0.0848	0.1698	0.0132	0.0010
Kupiec $_{POF}$	0.0237	0.0052	0.0579	0.0375
Christoffersen $_{CCI}$	0.8575	0.2998	0.7061	0.7810
Christoffersen $_{CC}$	0.0763	0.0117	0.1542	0.1106
KLM $_{mult}(4)$	0.0838	0.0088	0.0284	0.0017
Rejections / tested	5/11	8/11	3/9	7/9
Log-likelihood (IS)	−2860.23	−2915.35	−2936.60	−3002.10

In-sample, the four variants rank cleanly: GJR- t (−2860.23) dominates GARCH- t (−2915.35) by 55 log-points; the Student- t variants beat the Normal variants by a much larger margin. The

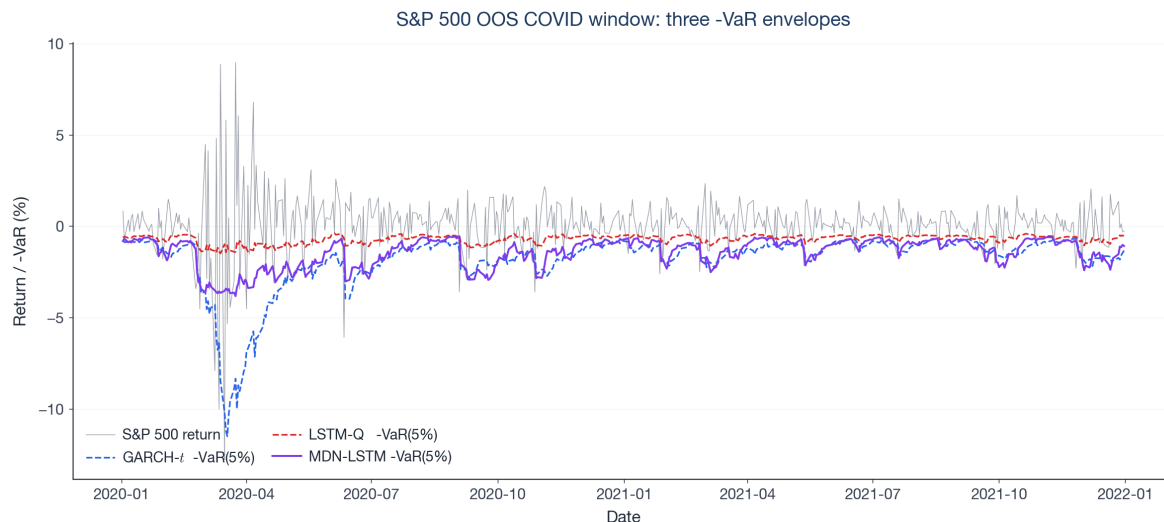


Figure 9: S&P 500 daily returns over the COVID-19 OOS slice and the three negative VaR envelopes ($\alpha = 0.05$): GARCH- t scales with the realised volatility regime, the LSTM-Quantile head holds approximately constant at the level of the calm 2010–2019 in-sample window, and the MDN-LSTM sits between the two.

order survives OOS: the rejection count drops from 8/11 for GARCH- t to 5/11 for GJR- t on the same data. The asymmetric variance recursion routes more variance toward days following negative shocks, which is exactly the clustering pattern $C_{ES}(m)$ and $C_{VaR}(m)$ detect; with $\gamma > 0$ the conditional Box-Pierce p-values move from 0.0164 to 0.2366 and from 0.0045 to 0.9899 respectively. The Acerbi–Szekely Z_1 test inverts: GJR- t rejects ($p = 0.014$) where GARCH- t does not ($p = 0.107$), because the leverage term predicts a smaller ES on the violation days and the realised severities exceed it.

The two Normal-innovation variants lose the AS p-values for the reason given in the table caption; their remaining tests show that $KLM_{\text{mult}}(4)$ rejects in both cases ($p < 0.05$), which is the multinomial substitute for the missing Acerbi–Szekely signal. The Normal-innovation variants are dominated in log-likelihood and miss the AS block, so they are not competitive forecasting choices for an OOS slice that contains a fat-tail event. Figure 10 renders the eleven-by-four p-value grid; variants are ordered by in-sample log-likelihood.

5 Discussion

The empirical results in Section 4 converge on three findings that have operational implications for ES backtesting.

Eleven tests give eleven answers. On the same out-of-sample slice, the unconditional and conditional Du–Escanciano statistics, the Kupiec proportion-of-failures, the Christoffersen decomposition, the Kratz–Lok–McNeil multinomial test, the Acerbi–Szekely Z_1 and Z_2 , and the McNeil–Frey bootstrap occupy different regions of the p-value space. On the S&P 500 GFC window the tests split nine to four; on the COVID window ten to three; the three tests that fail to reject on both windows (Christoffersen $_{CCI}$, Z_1 , McNeil–Frey) overlap. On the multi-asset grid the pattern is sharper: the marginal-coverage block reaches 19–20 of 20 non-SVB cells,

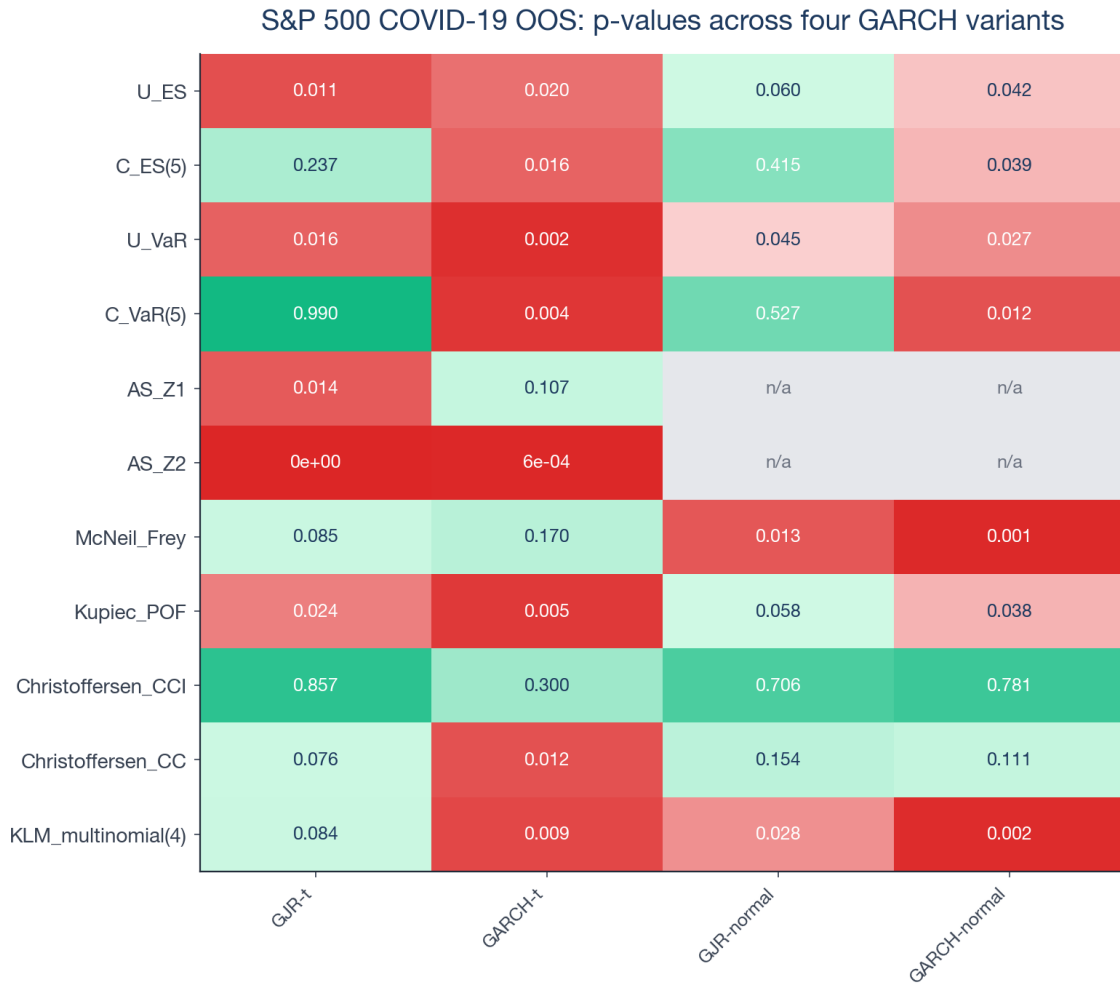


Figure 10: P-value heatmap across (test) $\times$ (variant) cells on the S&P 500 COVID-19 OOS slice. Cells with $p < 0.05$ are shaded red, cells failing to reject are shaded green; the intensity tracks distance from the threshold. Variants are ordered left-to-right by in-sample log-likelihood (best first): GJR- t , GARCH- t , GJR- $\mathcal{N}$, GARCH- $\mathcal{N}$.

the conditional Box–Pierce block reaches 7–9, Christoffersen_{CCI} reaches 2. The choice of test is therefore not a sensitivity check on the result; it is a choice of which channel of mis-specification the test is designed to see. The unconditional block sees the violation count, $C_{ES}(m)$ sees clustering in the cumulative severity, Z_2 sees the joint frequency-and-magnitude tail behaviour, Z_1 and McNeil–Frey see the severity conditional on a violation.

The Christoffersen decomposition is more informative than the joint test. The $LR_{CC} = LR_{POF} + LR_{IND}$ decomposition is exact. The empirical results assign almost the entire signal to LR_{POF} , which is also the simpler test. A practitioner running only LR_{CC} would observe a rejection without learning whether the failure comes from violation count or from clustering; the decomposition recovers that information at no additional computational cost. The case for reporting (LR_{POF}, LR_{IND}) rather than LR_{CC} is thus diagnostic, not statistical.

Z_2 dominates Z_1 under leverage and on real crises. The Monte Carlo study shows that Z_2 's net power under strong asymmetric leverage at $n = 1,000$ is +9 pp while Z_1 's is 0 pp. The same

hierarchy holds on real data: across the twenty non-SVB cells in the grid, Z_2 rejects in 19, Z_1 in 5. The two statistics are sometimes presented as alternatives among the “Acerbi–Szekely tests”; the empirical evidence here suggests treating them as different tests. A practitioner who wants a single ES-specific check that responds to leverage would pick Z_2 .

Marginal-coverage tests over-reject at moderate n . Under correctly-specified DGP the empirical size at $n = 1,000$ is 0.16 for U_{ES} and McNeil–Frey, 0.14 for U_{VaR} and Z_2 , 0.13 for Kupiec_{POF} and KLM_{mult}(4), and 0.12 for Z_1 and Christoffersen_{CC}. The asymptotic χ^2 and $\mathcal{N}(0, 1)$ distributions are reached only at substantially larger n . Calibration of operational backtest thresholds should account for this size distortion; quoting a 5% nominal threshold for U_{ES} at $n = 1,000$ overstates the type-I rate by a factor of roughly 3.2.

Neural baselines do not generalise across volatility regimes. The LSTM-Quantile head trained on 2010–2019 under-calls COVID-19 VaR by a factor of 2.6. Both models reject the Kupiec test, but the LSTM’s rejection is twenty times stronger than the GARCH’s. The architecture is not a substitute for an explicit volatility-clustering model when the OOS slice contains a regime that the training data does not.

The MDN-LSTM fixes the architectural gap, not the regime-shift one. Replacing the two-output head with three heads that parameterise the Student- t distribution supplies the PIT, the analytic VaR and ES, and the parametric simulator the LSTM-Quantile lacks. Eleven of the thirteen statistics now apply on the neural baseline. The MDN-LSTM also under-calls COVID-19 VaR (1.45% versus GARCH- t ’s 1.83%), but at a factor of 1.3 rather than 2.6 – the parametric output lets the neural network learn a fat-tailed conditional distribution from the in-sample window. The remaining gap is a training-data issue, not an architectural one.

The GARCH-variant choice is part of the backtest. Holding the OOS slice and the test battery fixed, swapping the symmetric GARCH(1,1) for the asymmetric GJR(1,1,1) flips five of the eleven verdicts (Table 10): the conditional Box-Pierce statistics on cumulative violations relax to non-rejection, the Christoffersen_{CC} statistic relaxes, the multinomial test relaxes. The Acerbi–Szekely Z_1 inverts in the opposite direction. The empirical evidence says that “which test” and “which model” are coupled choices that need to be reported together; the practice of fixing the forecasting model and reporting backtest results as a model-validation statement on its own is a half-question.

6 Conclusion

Expected shortfall as a regulatory risk measure has clean theoretical foundations and a fragmented backtesting literature. This paper implements eleven of the leading ES and VaR backtests in a single unified battery and shows that the choice of a single test materially shapes the verdict. On the S&P 500 the eleven tests disagree on three of the thirteen statistics on every crisis window; on a grid of five indices and five crises they disagree on twenty p-values out of twenty per non-trivial test. The Monte Carlo size-and-power study quantifies the under- and over-rejection

of each test at finite sample sizes and shows that Z_2 dominates Z_1 by a wide margin against asymmetric leverage. A four-variant sensitivity check isolates the contribution of the GARCH choice: switching from symmetric GARCH to asymmetric GJR flips five of the eleven verdicts on the same OOS data. A mixture-density LSTM extends the classical neural baseline so it can run on the full battery, cutting the COVID under-call from a factor of 2.6 to a factor of 1.3. The operational takeaway is that ES should not be backtested by a single statistic, and that the choice of forecasting model is part of the backtest rather than a separable input.

The implementation is a self-contained Python package (esback) released under the MIT licence; the architecture and the recipe to reproduce every number in this paper are documented in Appendices A and B.

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A Software architecture

The implementation is organised under `src/esback/`:

- `config.py`, `_logging.py`: project-wide constants (random seed, paths) and logging.
- `data/cache.py`, `data/loaders.py`: parquet cache around the `yfinance` downloader, with a frozen reference series under `tests/data/`.
- `models/ar1_garch_t.py`: baseline AR(1)-GARCH(1,1)- t fit and the shared `_garch_recursion` helper used by every variant.
- `models/garch_variants.py`: GJR-GARCH and Gaussian innovation variants routing through the shared recursion.
- `models/dl/`: the LSTM-Quantile head and its pinball + FZ0 loss functions, on top of PyTorch.
- `pit.py`, `risk_measures.py`: PIT residual transforms and closed-form VaR_t , ES_t for the standardised Student- t and Normal innovations.
- `backtests/du_escanciano.py`: the four basic and the two robust Du–Escanciano statistics.
- `backtests/var_baselines.py`: Kupiec POF, Christoffersen CCI and CC, Kratz–Lok–McNeil multinomial.
- `backtests/es_baselines.py`: Acerbi–Szekely Z_1 and Z_2 with their parametric simulator; the McNeil–Frey bootstrap.
- `backtests/battery.py`: unified `run_full_battery` entry point.
- `pipeline.py`, `multi_asset.py`, `simulation/`: single-period, multi-asset and Monte Carlo orchestration.

- `viz/style.py`, `viz/figures.py`: a single palette and `apply_rcparams` helper plus thirteen figure builders.
- `cli.py`: `esback replicate` and `esback multi-asset` entry points, wired through `typer`.

The unified battery is exposed as `run_full_battery(...) → dict[str, BacktestResult]`. Every `BacktestResult` carries the test name, the statistic, the p-value, the alpha at which the test was run, an asymptotic-null reference and a free-form extra dictionary for diagnostic quantities.

B Reproducibility

The package supports a deterministic end-to-end re-run on Python ≥ 3.11 . The recommended setup is

```
pip install -e ".[dev]"
pytest
```

followed by the two command-line entry points

```
esback replicate    -c configs/replication_du_escanciano.yaml
esback multi-asset  -c configs/multi_asset_crisis.yaml
```

which run in ~ 2 seconds and ~ 3 minutes on Apple Silicon respectively. The S&P 500 returns series is pinned to a parquet file under `tests/data/`; the regression test suite under `tests/integration/test_regression.py` asserts that PIT residuals, VaR and ES paths, and every backtest p-value match the committed reference outputs to 10^{-10} on path quantities and to 10^{-12} on p-values. The Monte Carlo study results are cached to `results/power_study/results.parquet`; a refresh is triggered by deleting that file.

The package is style- and type-checked under `ruff` and `mypy`; the unit and integration suites cover 181 tests that run in under ten seconds. The notebooks under `notebooks/` are executable from a clean checkout via `jupyter nbconvert -to notebook -execute -inplace`; the six notebooks regenerate the ten figures bundled under `assets/figures/`.